

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	
Team ID	
Project Name	Credit Card Approval Prediction
Maximum marks	

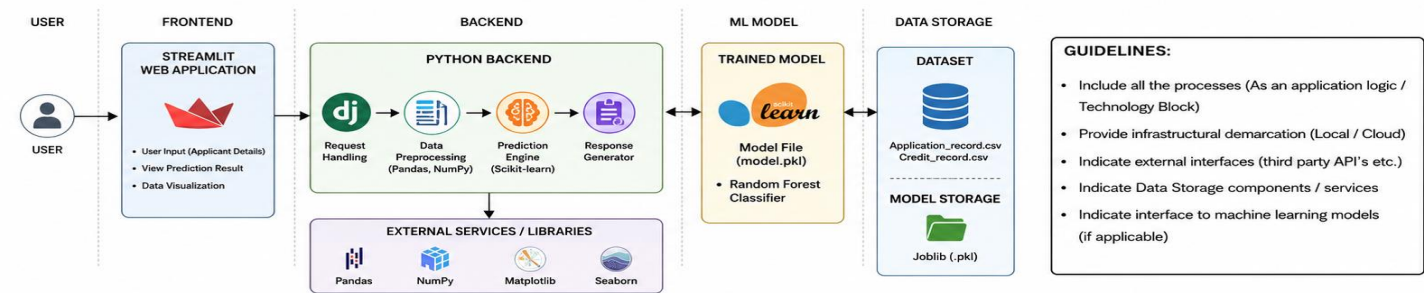
Technology Stack (Architecture & Stack)

Introduction

The **Credit Card Approval Prediction System** is developed using Python and Machine Learning. It predicts whether a credit card application will be approved or rejected based on the applicant's information and credit history.

Technology Stack

Component	Technology Used	Purpose
Frontend	Streamlit	User Interface
Backend	Python	Application Logic
Machine Learning	Scikit-learn	Model Training and Prediction
Data Processing	Pandas, NumPy	Data Cleaning and Analysis
Data Visualization	Matplotlib, Seaborn	Charts and Graphs
Model Storage	Joblib	Save and Load Model
Development Tool	VS Code, Jupyter Notebook	Coding and Testing
Version Control	Git, GitHub	Source Code Management



S.No	Component	Description	Technology
1.	User Interface	Allows user to enter applicant details and view prediction result.	Streamlit, HTML, CSS, JavaScript
2.	Python Backend	Handles application logic, request processing and response generation.	Python 3.x
3.	Data Preprocessing	Cleans, transforms and prepares input data for model prediction.	Pandas, NumPy
4.	Prediction Engine	Loads the trained model and predicts approval / rejection.	Scikit-learn
5.	Trained ML Model	Machine learning model used for prediction.	Random Forest Classifier (model.pkl)
6.	Dataset	Contains applicant information and credit history.	CSV Files (Application_record.csv, Credit_record.csv)
7.	Model Storage	Stores the trained machine learning model file.	Joblib (.pkl)
8.	Data Visualization	Used for EDA and result visualization.	Matplotlib, Seaborn
9.	External Libraries	Support libraries for computation and analysis.	Pandas, NumPy, Matplotlib, Seaborn

Advantages

- Easy to use.
- Fast prediction process.
- Accurate machine learning model.
- User-friendly web interface.
- Reduces manual effort in credit card approval.

Conclusion

This technology stack provides a simple, efficient, and reliable solution for predicting credit card approval. It combines data processing, machine learning, and a user-friendly interface to improve the decision-making process.